

Exploring the Negative Effects of Gamification on Students: A PRISMA-Based Systematic Literature Review

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Abstract—Gamification has been widely used in education as a solution to increase student motivation and interest in learning. Previous studies have examined the benefits of gamification and its core elements. However, few studies have investigated the negative effects of gamification elements. This study conducted a systematic literature review (SLR) in accordance with PRISMA 2020 guidelines to examine the negative impact of gamification elements on students. A total of 206 studies published between 2020 and 2025 in seven major databases were identified, of which 20 were selected based on inclusion and exclusion criteria. The results show that the negative impacts of gamification can be divided into four categories: performance pressure and psychological strain, superficial learning effects, motivational engagement decline syndrome, and psychological distress and emotional strain. Gamification elements such as leaderboards, challenges, points, narratives, and rewards can weaken students' basic psychological needs: autonomy, competence, and relatedness if not well designed and implemented. Gamification can shift students' focus from primary learning to competition, achievement, and performance demands. Educational institutions seeking to develop and implement gamification in their learning systems must select elements aligned with learning objectives and student characteristics, while considering potential impacts. Thus, wisely selecting gamification elements can support the learning process and increase student engagement.

Keywords—gamification, gamification elements, negative effect, learning motivation, systematic literature review

I. INTRODUCTION

Digital learning is increasingly being used as an innovative learning tool, one of the applications of digital learning is gamification, gamification is present as a solution to increase student motivation and involvement in the learning process. Gamification features have been widely implemented in digital learning platforms, such as Kahoot and Quizizz. They popularize gamification elements such as points, challenges, and leaderboards to improve learning outcomes, thereby encouraging educational institutions to adopt similar practices. This widespread use has created an implicit narrative that gamification is universally beneficial; empirical evidence across contexts shows otherwise.

This study examines a different perspective on the positive stigma of gamification, revealing the potential dangers of its implementation without pedagogical alignment. Previous research has reported that some students may perceive gamification as distracting, stressful, and demotivating [1]. The implementation of gamification elements still produces superficial, purely cosmetic effects, such as points and progress features that are less meaningful [2]. These findings reinforce that gamification is not entirely neutral; poor design can have significant negative impacts.

Prioritizing learning and intrinsic motivation can be diminished if gamification relies too heavily on extrinsic incentives. Leaderboards and other competitive elements can lead to poor social communication, creating anxiety and potentially lowering students' self-esteem. We'll review 10 key elements of gamification. Complementing our observations of more than 20 gamification risks, this pattern indicates that gamification has a substantial dark side [3].

Despite growing awareness of this issue, there appears to be little systematic study of it. This results in fragmented knowledge. To address this risk, this study aims to systematically categorize the dark side of gamification elements and clarify how misaligned design decisions contribute to it. In doing so, this perspective seeks to support more responsible, evidence-based, and learner-sensitive gamification practices.

II. LITERATURE REVIEW

This literature review examines four key aspects of gamification in education. The review begins with an explanation of gamification as a general concept, followed by its context in education. Next, the review discusses the elements of gamification. Finally, the negative impacts of gamification, highlighting the potential weaknesses and consequences identified in previous studies.

A. Gamification

Gamification refers to the use of game elements in non-game contexts, such as education. Gamification is widely used to increase student interest and learning outcomes. Game

elements are incorporated into the learning process to create a competitive yet enjoyable learning environment for students [4].

B. Gamification Elements

Gamification elements are the components that constitute the gamification process. Examples of these components include leaderboards, badges, points, rewards, and many more. In education, gamification elements are used to increase student interest in continuing learning and motivation [5].

C. Gamification in Education

Selecting elements that align with students' needs can help educational developers design innovative and effective gamification strategies [5]. Many gamification applications, such as Kahoot and Quizizz, have been implemented on online learning platforms, both of which have been shown to increase student engagement and enthusiasm, thereby encouraging more frequent participation in learning activities. The effectiveness of game components in learning can be maximized when they are selected and designed to align with learning objectives and students' needs, while accounting for the positive and negative effects of each component [6].

D. Negative Effect of Gamification

Gamification has been shown to increase student engagement in learning and intrinsic motivation [7]. The application of game elements in the learning process can enhance the learning experience, making the learning environment more interactive [8]. However, it is essential to note that gamification also has negative effects; gamification-based learning does not encourage creative thinking and limits student activity [9]. The use of Points and Leaderboards elements can cause negative emotions in students when their rankings are low, leading to discomfort and a decline in motivation over time [10]. On the other hand, elements such as rewards can lead students to focus more on achieving them than on learning the core material [11].

III. RESEARCH METHODOLOGY

This SLR was conducted using the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) methodology, which is widely used, transparent, and structured [12].

A. Eligibility Criteria

Eligibility criteria are essential to ensure that all reviews are valid, relevant, and cover the required research area. If an article meets the first exclusion criterion, it is immediately excluded and does not require rechecking against the subsequent exclusion criteria. The following exclusion criteria were applied:

- Article Published before 2020
- Duplicate publications found in multiple databases
- The article with incomplete or insufficient data to support the analysis
- Articles not written in English

B. Search Strategy

The literature search was conducted in seven academic databases: AIS Electronic Library, ACM Digital Library, IEEE Xplore, ScienceDirect, Springer, Taylor & Francis

Online, and Wiley Online Library. These databases were selected because they provide credible publications in the fields of education and technology. Keywords were structured using the Boolean operators AND and OR to ensure more relevant and comprehensive search results. The three search strings used were:

- Gamification AND (education OR learning) AND (factor OR component) AND (motivation OR engagement) AND (effectiveness OR impact).
- Gamification AND (design OR framework) AND (element OR principle) AND (education or e-learning) AND (outcome OR evaluation).
- Gamification AND (student OR learners) AND (mechanics or elements) AND (performance OR achievements) AND (education or training).

All retrieved articles were then processed using Mendeley Reference Manager to remove duplicates and were selected based on inclusion and exclusion criteria.

C. Study Selection

The literature selection process follows the PRISMA 2020 standard flow, comprising four main stages: identification, screening, eligibility, and inclusion.

- Identification: This stage involves collecting studies from 7 major databases and removing duplicates using Mendeley Data (version 2.1.3.9).
- Screening: The selection stage for titles and abstracts that do not meet the exclusion criteria, defined as studies published before 2020 that do not discuss gamification or the impact of gamification elements.
- Eligibility: At this stage, studies that pass the selection process are reviewed in full (full-text review). Studies with incomplete or insufficient data to support the analysis will be eliminated.
- Inclusion: Studies that meet the inclusion and exclusion criteria are included in the systematic literature review; the analysis of results will be based on these studies.

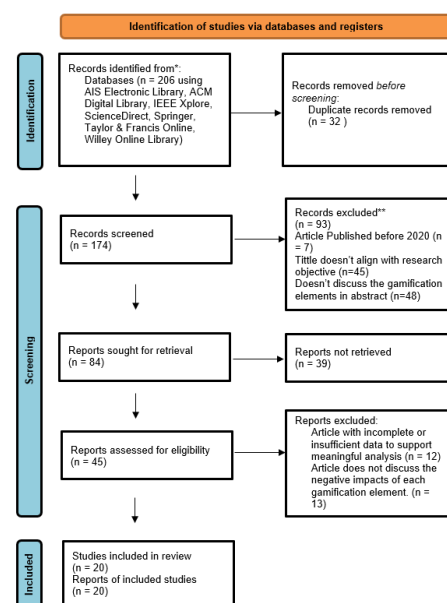


Fig. 1. PRISMA 2020 Flow Diagram for New Systematic Review

Figure 1 describes the study selection process. The study selection process uses the PRISMA 2020 framework. The selection stage comprises three stages: identification, screening, and inclusion. In the identification stage, 206 articles were retrieved from 7 major databases: AIS Electronic Library, ACM Digital Library, IEEE Xplore, ScienceDirect, Springer, Taylor & Francis Online, and Wiley Online Library. Subsequently, 32 articles were excluded as duplicates, leaving 174 articles that proceeded to the screening stage. In the screening stage, 93 articles were eliminated because they were published before 2020 and did not discuss gamification or its elements in education.

Furthermore, 84 articles were read in full, but 39 articles were eliminated. The remaining 45 articles, 45 articles were then evaluated at the feasibility stage. A total of 25 articles were eliminated because 12 were inadequate for meaningful analysis, and 13 did not discuss the negative impacts of each gamification element. In the end, 20 articles that met all inclusion and exclusion criteria were included in the analysis for this SLR.

D. Information Regarding The Selected Study

Among the 20 selected studies, the most significant proportion comes from the Educational Technology discipline (35%), followed by Information Systems (25%) and Computer Science (10%). Additionally, Information and Software Technology represents 5%. Outside of information technology, we selected studies in the fields of Psychology (5%), Medical Education (5%), and Physical Education Teacher Education (5%). The distribution of studies across disciplines is visually summarized in Figure 2.

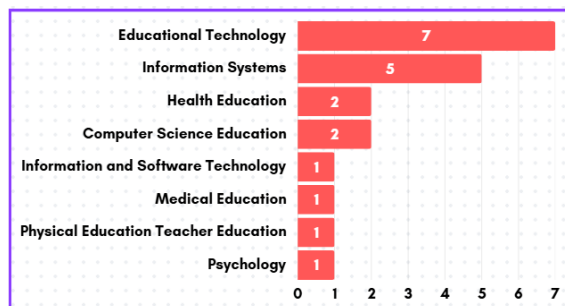


Fig. 2. Distribution of Fields in Selection Studies

Figure 2 illustrates that research on gamification in education is highly interdisciplinary, with applications across nearly every field of educational study. The studies selected for this research were drawn from journal articles and conference proceedings. Table I shows the source of each study, whether published in a journal or conference, which also illustrates the latest developments in this field.

TABLE I. SOURCE AND PUBLICATION OF SELECTED STUDIES

Reference No.	Year	Source	Type
[2]	2023	International Journal of Changes in Education	Journal
[3]	2022	Journal Information and Software Technology	Journal
[13]	2023	Smart Learning Environments	Journal
[14]	2025	Journal of Education and Information Technologies	Journal

Reference No.	Year	Source	Type
[15]	2021	2021 44th International Convention on Information, Communication and Electronic Technology (MIPRO)	Conference
[16]	2024	Proceedings - 2024 9th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference, SEEDA-CECNSM 2024	Conference
[17]	2024	2024 3rd International Conference on Creative Communication and Innovative Technology, ICCIT 2024	Conference
[18]	2023	Proceedings of the 56th Hawaii International Conference on System Sciences	Conference
[19]	2024	Journal of Prevention	Journal
[20]	2021	Interactive Learning Environments	Journal
[8]	2020	IEEE Access	Journal
[21]	2020	IEEE Access	Journal
[22]	2024	Cogent Education (2024)	Journal
[23]	2022	Clinical Anatomy (2022)	Journal
[10]	2025	British Journal of Educational Technology	Journal
[24]	2023	International Journal of Environmental Research and Public Health	Journal
[25]	2025	ACE 2025 - Proceedings of the 27th Australasian Computing Education Conference, Held in conjunction with: Australasian Computer Science Week	Conference
[26]	2023	International Journal of Environmental Research and Public Health	Conference
[11]	2025	Acta Psychologica	Journal
[27]	2024	International Journal of Educational Research	Journal

IV. RESULT AND DISCUSSION

This analysis begins by describing the function of each element in each selected study. This is followed by a discussion of the negative effects of each gamification element, based on existing studies, and then divides these effects into four main categories. Self-Determination Theory (SDT) is used as an analytical lens to help understand the emergence of these negative impacts.

Based on the 20 studies analyzed, 10 gamification elements were identified. These elements include points, badges, leaderboards, feedback, challenges, storylines, progress bars, rewards, levels, and avatars. Before discussing the findings on the negative effects of gamification, the author outlines the definitions of each gamification element and the indicators used to measure them. Table II outlines ten gamification elements frequently used in gamification development. Each of these elements has a different function.

TABLE II. DEFINITION AND INDICATOR OF GAMIFICATION ELEMENTS

Element	Definition	Indicator
Points	Points are a common element in gamification, usually their function is to provide information to students regarding the score for the work they have completed, this element can be a reference or measure of how well students have mastered the material given [8], [14], [15].	Measure of achievement
		Performance feedback
		Progress indicator

Element	Definition	Indicator
Badges	Badges can be used as awards for students who complete certain stages. Typically, the higher the level or challenge, the more prestigious the badges can be. They serve to increase student motivation and engagement [8], [14], [15].	Visual representation of achievement
		Sense of accomplishment
		Recognition
Leaderboard	The leaderboard displays a list of student information and the scores of competing students. This element shows a student's performance compared to other students. A competitive yet healthy environment will increase student motivation and engagement [8], [15]	Trigger competition
		Peer performance comparison
		Social motivation
Feedback	Feedback provides explanatory information about the tasks that students have completed, helps evaluate errors, helps progress, and maintains motivation. Teachers here also play a role in explaining as clearly as possible and maintaining student motivation [14], [15], [20].	Instant feedback
		Error correction guidance
		Progress monitoring
Challenges	Challenges provide tasks with varying levels of difficulty according to the level of the participants, encouraging problem solving and increasing engagement, but challenges that are too difficult will cause frustration [8], [19].	Balanced difficulty
		Problem-solving
		Engagement
Storyline	Storylines provide narrative context for learning experiences that are relevant to the real world, enabling students to connect learning materials to real cases, and increasing students' long-term engagement [15], [19].	Immersion
		Connection to real life context
		Emotional motivation
Progress Bar	Progress bar provides visual content of student progress, and the distance between their progress and learning targets, by providing transparency it can increase engagement [14], [16], [20]	Visual progress tracking
		Transparency of achievement
		Motivation to continue
Rewards	Rewards function as a reinforcement of motivation for student achievement, their forms can vary according to the purpose of the reward itself, this element increases student motivation and involvement [15], [16], [18].	External incentives
		Positive reinforcement
		Participation driver
Levels	Levels divide participants into different levels of difficulty and material; levels create a sense of progress and directed mastery of the material.	Progressive difficulty
		Progress recognition
		Mastery milestone
Avatars	Avatars are digital characters that students can customize to represent themselves in gamification, creating personalization and increasing engagement. [14], [15]	Identity personalization
		Flexibility of self-expression
		Social engagement

This table summarizes the functions and definitions of the 10 elements discussed in this paper, to clarify the concept. These elements are grouped into three categories: Competitive Elements, Reward/Feedback Elements, and Contextual Elements. Competitive elements consist of leaderboards and challenges, which emphasize performance comparison and problem-solving difficulty levels to stimulate student engagement and progress during the learning process. Reward/Feedback elements include points, badges, feedback, and rewards, which function to provide recognition for students and feedback during the learning process, which will

stimulate learners to remain engaged and progress. Finally, contextual elements comprise storylines, progress bars, levels, and avatars that support meaningful, personal, and progressive learning experiences. This classification highlights the different functional roles of gamification elements in shaping students' learning experiences.

The Negative effects of gamification are divided into four main categories. First, performance pressure and psychological strain refer to the pressure to excel or maintain achievements that trigger anxiety, insecurity, and a shift in focus from learning to performance demands. Second, the superficial learning effect refers to a condition in which gamification elements serve only as cosmetic features without instructional value, encouraging students to focus on points, badges, or rewards rather than on understanding the material in depth. Third, motivational engagement decline syndrome refers to a condition in which gamification elements reduce student motivation and engagement due to excessive challenges, confusing feedback, or unhealthy competition. Lastly, psychological distress and emotional strain refer to the emotional pressure that arises when gamification elements cause stress, anxiety, and frustration, making students feel mentally burdened. Therefore, to further elucidate these negative impacts, this study adopts self-determination theory as its lens.

A. Self-Determination Theory as a Theoretical Lens

Self-Determination Theory (SDT) provides a theoretical framework for understanding how human motivation works, emphasizing three basic psychological needs: autonomy, competence, and social relatedness. According to SDT, learning environments that support these needs tend to foster intrinsic motivation, while environments that undermine these needs in the context of gamification, such as excessive competition, inappropriate extrinsic rewards, or rigid performance metrics, can inadvertently interfere with psychological needs or even undermine them. Previous studies have reported various undesirable outcomes of gamification, including increased performance pressure, superficial engagement in tasks, decreased engagement, and increased psychological distress such as anxiety and loss of self-confidence. The four categories of negative effects of implementing gamification elements arise when gamification fails to support and disrupts basic psychological needs according to Self-Determination Theory. Therefore, SDT is an appropriate theoretical framework for categorizing the negative impacts of gamification.

B. Performance Pressure and Psychological Strain

Negative impact of gamification is the emergence of pressure. Students who feel pressured will lose focus on the primary learning material. For example, avatars can make students feel stressed and distract them from the core material [15]. When students choose their avatars, they often seek to appear cooler than their peers, which creates social pressure because they fear being compared or judged as inferior. The effect this has is that students' focus shifts from learning to having the best avatar.

The second element is challenge. Challenge elements, such as competition and time tracking, can put pressure on students and distract them from their primary learning

objectives [15], [21]. Thus, students feel pressured because they fear losing to their peers, which aligns with the leaderboard element. Leaderboards can motivate students to participate more actively, but they can also create pressure when scores are low or positions are lower [14]. For this reason, relative leaderboards have been proposed as a potential solution to reduce this problem [24]. Relative leaderboards display a student's position relative to a small group or a few people near their rank (e.g., two above and two below). The purpose is to reduce pressure and avoid embarrassing students who rank at the bottom.

Most respondents in the study reported that the reward system can create pressure and anxiety [19]. Other studies also mention that poorly designed reward elements can make students feel pressured [8], [18]. The pressure referred to here is excessive pressure to perform perfectly. If the reward system does not adapt to students' needs or development, it can damage their internal motivation and cause feelings of helplessness and pressure [8], [18].

C. Superficial Learning Effect

Prior research indicates that certain students perceive badges as lacking meaningful pedagogical value. This occurs when badges serve only cosmetic purposes without a direct connection to learning objectives. [23]. Narratives or stories in gamification can affect student engagement if the storyline is not relevant to the material; students may view it as a gimmick, diverting attention from the core concepts. [2], [13], [21].

Starting from points created solely to provide external rewards can produce a greater sense of validation and increase students' intrinsic motivation [27]. On the other hand, points and levels are under pressure, which makes students feel less free to explore their learning progress; instead, they focus on pursuing rewards [8]. The effectiveness of the points element was not experienced uniformly by students: 10.1% of respondents reported that the score was not at all helpful, and only 13.1% reported that the aspect was beneficial [20]. When points do not provide clear instructions, students tend to ignore them and treat them as merely cosmetic, while the reward element can have detrimental consequences if over-exploited; the focus shifts toward obtaining the reward at the expense of shallow engagement [14]. In some cases, overreliance on rewards can lead to disengagement, in which learners focus on pursuing rewards rather than on acquiring knowledge from the learning material [3].

D. Motivational-Engagement Decline Syndrome

Category of effects that can result from the implementation of gamification in education is the motivational engagement decline syndrome, which refers to gamification elements that reduce students' motivation and interest in learning. One of the influential elements in this case is the challenge. If challenges do not align with students' abilities, they may feel confused and overwhelmed, ultimately leading to a loss of interest in learning. Excessive or overly complex challenges can create stress and decrease student interest [15], [21].

Feedback also negatively affects student motivation and engagement. Excessive feedback can confuse students; rather than being beneficial, it can reduce their interest in learning by overwhelming them. Furthermore, feedback that doesn't provide clear direction can increase competitiveness among students, thus leading to a syndrome of demotivation and engagement [18], [21].

Leaderboards can have negative effects if competition becomes less supportive [18], [21]. Leaderboards can damage social relationships among students at school by encouraging excessive competitiveness, particularly when students' rankings are compared with those of higher-ranking students. Other research has found that competitive elements, such as leaderboards, can have negative effects, including decreased interest in learning among students who are ranked low or who do not appear on the leaderboard [22]. Why is this? When students' self-confidence is lost, they feel like failures, which ultimately leads to frustration and giving up, including refusing to continue learning.

Furthermore, the leaderboard creates pressure for students to maintain their rankings. This pressure can lead to demotivation, particularly among students in the low-ranking range. Being in a low position can lead students to feel they have failed to complete the task, which, in turn, can trigger a loss of self-confidence in their learning abilities (self-efficacy) [25]. Additionally, when they are in the low-ranking range, negative emotions arise within students. Feelings of discomfort and decreased motivation to learn are also present in this situation [10].

Rewards can also negatively impact student motivation and engagement. Rewards that are inappropriate for age, ability, and achievement levels can trigger a decline in motivation [22]. Rewards that are irrelevant to age or childlike behavior can make students feel less appreciated academically, making them uncomfortable and reluctant to learn. Rewards design that takes student characteristics into account is necessary to prevent disengagement [13]. Beyond increasing students' interest and motivation to continue learning, rewards can also lead students to focus more on obtaining the reward than on the core material. This, of course, leads to shallow involvement and reduces learning outcomes [11]. The final element of gamification that impacts interest and motivation is narrative. Narratives that are childish and not based on reality will leave students bored because the story is not appropriate for their age [26]. Therefore, the narrative must be age-appropriate to maintain students' engagement throughout the learning process.

E. Psychological Distress and Emotional Strain

Gamification can also trigger psychological distress and emotional strain, which refers to emotional pressure such as stress, anxiety, and frustration, thus burdening students mentally. One element of gamification that can trigger stress and frustration is challenge. Research shows that when a challenge is too difficult, students may become frustrated because they feel they cannot complete it, thereby decreasing their motivation to continue learning [15], [21]. If students repeatedly fail to complete assigned challenges, they may become frustrated, which leads to stress and feelings of helplessness [18]. In addition to rewards, feedback can trigger

stress. Feedback that is not delivered clearly or is long-winded can trigger anxiety and stress, which ultimately makes students feel uninvolved during the learning process [18], [21].

Points and leaderboards can also trigger anxiety and stress if students are compared and evaluated on their achievements with others [10]. A study reported that some students felt anxious due to the leaderboard system in gamification [19]. The use of a level structure, which was created for games, when applied to learning tasks, can make students feel socially and mentally overwhelmed (social overload and strain) [17]. In gamification, level systems typically require players to progress quickly through levels or to demonstrate their achievements relative to others. This pattern is applied in educational or work contexts, users may feel pressured to continually improve their performance, be constantly socially compared, and experience mental fatigue due to demands for progress that are not always aligned with their actual learning. Students who rank low may feel inferior and left behind, which can damage their self-esteem and disrupt the classroom's collaborative environment [27]. Finally, progress bars that set overly high targets can make students feel overwhelmed and stressed [21]. They see their own progress as slow, far behind others, or the targets that must be achieved.

V. CONCLUSION

This review investigated the negative implications of gamification elements in educational settings and found that negative effects frequently arise when gamification is implemented without a firm pedagogical foundation or alignment with learners' needs. The findings highlight recurring issues such as heightened performance pressure, anxiety, and a redirection of students' focus from learning goals toward competition and external recognition. Several commonly used gamification elements, including points, badges, rewards, leaderboards, and narratives, may lead to surface-level participation and weaken intrinsic motivation when they function primarily as decorative features rather than meaningful learning tools. Furthermore, overly demanding challenges, rewards that do not align with learners' characteristics, and feedback that lacks clarity or direction can diminish learners' interest in learning and contribute to stress and frustration. Overall, the results suggest that gamification should not be viewed as a universal solution for increasing engagement or motivation. When poorly designed, it may fail to support learning and instead generate unfavorable effects on students' attitudes and behaviors. Consequently, effective gamification requires deliberate design choices, careful selection of elements, and consideration of both their positive contributions and potential drawbacks, along with strategies to reduce unintended negative outcomes.

IMPLICATION

This study can inform educational institutions considering the implementation of gamification in their learning systems. Gamification integration isn't always straightforward and requires careful evaluation and selection of the elements to be used. The design of each component must be carefully considered to avoid negative effects that could distract students from the primary learning objective and diminish

their interest in learning. Poorly designed gamification elements can lead to negative social dynamics among students [27].

FUTURE RESEARCH & LIMITATIONS OF THE STUDY

Further research is recommended to examine the impact of gamification elements in a more specific study area, to understand how the implementation of elements such as badges, leaderboards, avatars, points, etc. impacts students in each area of study, considering that differences in educational levels and educational majors have the potential to produce diverse findings. This research has not yet examined variations in the effects of these elements across cultural contexts; for example, in collectivistic cultures, individual competition can reduce engagement, while in competitive cultures, rewards can increase motivation. Therefore, it is essential to examine whether the negative effects of gamification vary across cultures or educational systems, and how cultural values may shape students' perceptions of competition and external rewards.

This study examines the negative effects of the gamification elements identified in the selected studies. This study does not discuss the negative effects caused by the combination of elements, for example, leaderboards with rewards and points, and what the negative impacts are when combined. This study is limited to investigating the negative effects of gamification in general across various levels of education, rather than in specific subject areas.

AUTHOR CONTRIBUTION

Muhammad Adhiim Nugroho was responsible for conceptualizing the idea, developing the background, collecting and analyzing the data, and drafting the original manuscript. Afina Salma contributed to the provision of research resources and the analysis of results and assisted with formatting the paper. Akmal Rasyid Pradipa was involved in the literature review, assisted with data collection and analysis, drafted the original manuscript, and validated the results. Edi Purnomo Putra supervised the overall study and provided critical feedback throughout the research process. All authors approved the final manuscript.

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